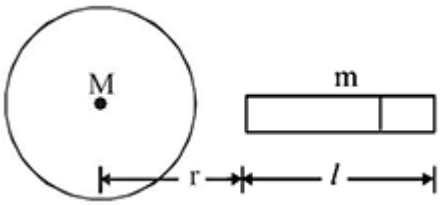
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The gravitational force of attraction between a uniform sphere of mass $\mathrm{M}$ and a uniform rod of length $l$ and mass $\mathrm{m}$ oriented as shown is




**(A)** $\frac{\mathrm{GMm}}{\mathrm{r}(\mathrm{r}+l)}$

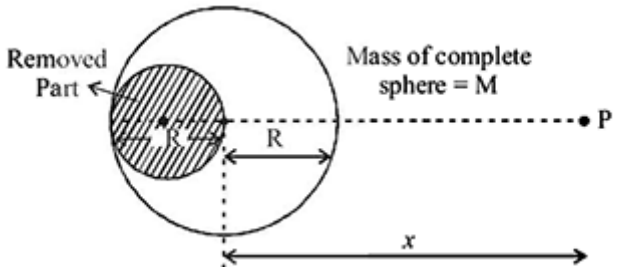
**(B)** $\frac{GM}{r^2}$

**(C)** $\mathrm{Mmr}^2+l$

**(D)** $\left(\mathrm{r}^2+l\right)\mathrm{mM}$

**Q.2**

The gravitational field, due to the 'left over part' of a uniform sphere (from which a part as shown, has been 'removed out'), at a very far off point, P, located as shown, would be (nearly):




**(A)** $\frac{5}{6}\frac{GM}{x^2}$

**(B)** $\frac{8}{9}\frac{GM}{x^2}$

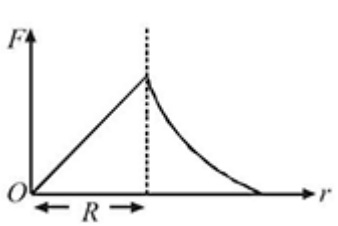
**(C)** $\frac{7}{8}\frac{GM}{x^2}$

**(D)** $\frac{6}{7}\frac{GM}{x^2}$

**Q.3**

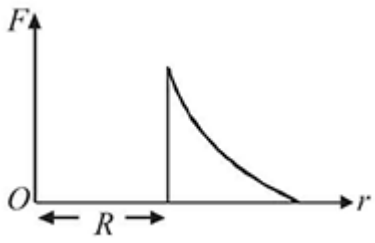
Which one of the following plots represents the variation of gravitational field on a particle with distance $\mathrm{r}$ due to a thin spherical shell of radius $\mathrm{R}?(\mathrm{r}$ is measured from the centre of the spherical shell)

**(A)**

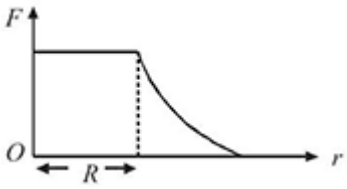



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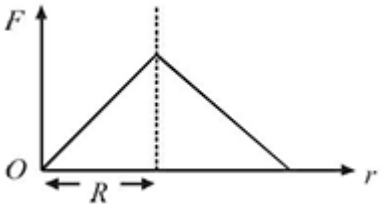
**(B)**




**(C)**




**(D)**




**Q.4**

In a gravitational field, at a point where the gravitational potential is zero, given that reference point is at infinity.

**(A)** the gravitational field is necessarily zero

**(B)** the gravitational field is not necesarily zero

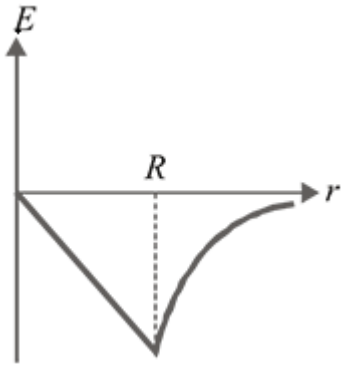
**(C)** any value between one and infinite

**(D)** None of these

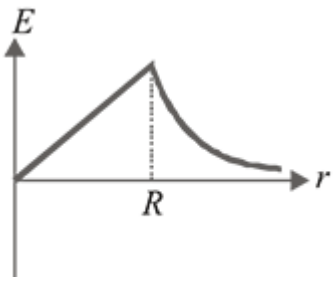
**Q.5**

The gravitational field strength due to a solid sphere (mass $M$, radius $R$) varies with distance $r$ from centre as

**(A)**

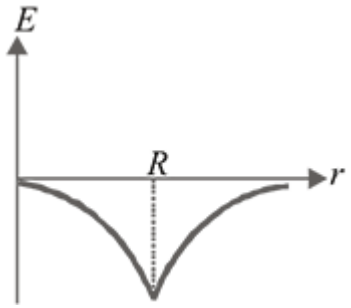



**(B)**

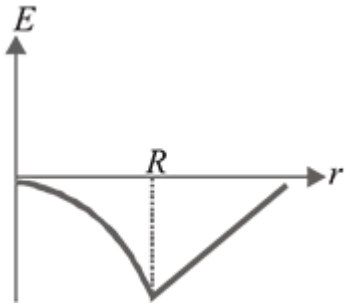



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**(C)**




**(D)**




**Q.6**

Two particles of equal mass ' $m$ ' go around a circle of radius $R$ under the action of their mutual gravitational attraction. The speed of each particle with respect to their centre of mass is

**(A)** $\sqrt{\frac{Gm}{4R}}$

**(B)** $\sqrt{\frac{Gm}{3R}}$

**(C)** $\sqrt{\frac{Gm}{2R}}$

**(D)** $\sqrt{\frac{Gm}{R}}$

**Q.7**

A straight rod of length $L$ extends from $x = a$ to $x = L + a$. Find the gravitational force it exerts on a point mass $m$ at $x = 0$ if the linear density of rod $\mathbf{y} = A + Bx^2$.

**(A)** $\mathrm{G}m\left[\frac{A}{a} + BL\right]$

**(B)** $\mathrm{G}m\left[A\left(\frac{1}{a} - \frac{1}{a+L}\right) + BL\right]$

**(C)** $Gm\left[BL + \frac{A}{a+L}\right]$

**(D)** $\mathrm{G}m\left[BL - \frac{A}{a}\right]$

**Q.8**

The acceleration due to gravity on the planet $\mathrm{A}$ is 9 times the acceleration due to gravity on planet B. A man jumps to a height of $2\ \mathrm{m}$ on the surface of $A$. What is the height of jump by the same person on the planet $\mathrm{B}$ ?

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**(A)** $\frac{2}{3}$ m

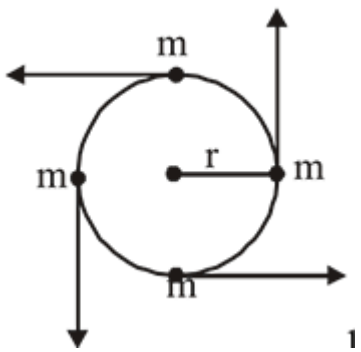
**(B)** $\frac{2}{9}$ m

**(C)** 18 m

**(D)** 6 m

**Q.9**

Four similar particles of mass m are orbiting in a circle of radius r in the same angular direction because of their mutual gravitational attractive force. Then, velocity of a particle is given by




**(A)** $\left[\frac{Gm}{r}\left(\frac{1+2\sqrt{2}}{4}\right)\right]^{\frac{1}{2}}$

**(B)** $\sqrt{\frac{Gm}{r}}$

**(C)** $\sqrt{\frac{Gm}{r}(1+2\sqrt{2})}$

**(D)** $\frac{1}{2}\left[\frac{Gm}{r}\left(\frac{1+2\sqrt{2}}{2}\right)\right]^{\frac{1}{2}}$

**Q.10**

Consider two solid uniform spherical objects of the same density $\rho$. One has radius R and the other has radius 2R. They are in outer space where the gravitational field from other objects are negligible. If they are arranged with their surface touching, what is the contact force between the objects due to their traditional attraction?

**(A)** $G\pi^2R^4$

**(B)** $\frac{128}{81}G\pi^2R^4\rho^2$

**(C)** $\frac{128}{81}G\pi^2$

**(D)** $\frac{128}{87}\pi^2R^4G$

**Q.11**

Two spherical bodies of mass $M$ and $5M$ & radii $R$ & $2R$ respectively are released in free space with initial separation between their centres equal to 12 $R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before collision is

**(A)** $2.5R$

**(B)** $4.5R$

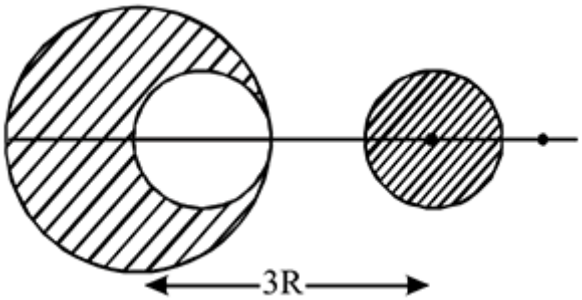
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**(C)** $7.5R$

**(D)** $1.5R$

**Q.12**

From a sphere of mass $\mathrm{M}$ and radius $\mathrm{R}$, a smaller sphere of radius $\frac{\mathrm{R}}{2}$ is carved out such that the cavity made in the original sphere is between its centre and the periphery (See figure). For the configuration in the figure where the distance between the centre of the original sphere and the removed sphere is $3\mathrm{R}$, the gravitational force between the two sphere is




**(A)** $\frac{41\mathrm{GM}^2}{3600\mathrm{R}^2}$

**(B)** $\frac{41\mathrm{GM}^2}{450\mathrm{R}^2}$

**(C)** $\frac{59\mathrm{GM}^2}{450\mathrm{R}^2}$

**(D)** $\frac{\mathrm{GM}^2}{225\mathrm{R}^2}$

**Q.13**

The value of 'g' at a particular point is $9.8\ \mathrm{m/s^2}$. Suppose the earth suddenly shrinks uniformly to half its present size without losing any mass. The value of ' $g$ ' at the same point (assuming that the distance of the point from the centre of the earth does not shrink) will now be

**(A)** $4.9\ \mathrm{m/sec^2}$

**(B)** $3.1\ \mathrm{m/sec^2}$

**(C)** $9.8\ \mathrm{m/sec^2}$

**(D)** $19.6\ \mathrm{m/sec^2}$

**Q.14**

While approaching a planet circling a distant star, an astronaut determines the radius of planet is half of that of the earth. After landing on its surface, he finds its acceleration due to gravity is twice as that of the earth. The ratio of the mass of planet and that of the earth.

**(A)** $1:2$

**(B)** $2:3$

**(C)** $3:4$

**(D)** $4:5$

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**Q.15**

The radii of two planets are respectively $\mathrm{R}_1$ and $\mathrm{R}_2$ and their densities are respectively $\rho_1$ and $\rho_2$ The ratio of the accelerations due to gravity at their surfaces is

**(A)** $g_1 : g_2 = \frac{\rho_1}{R_1^2} : \frac{\rho_2}{R_2^2}$

**(B)** $g_1 : g_2 = R_1 R_2 : \rho_1 \rho_2$

**(C)** $g_1 : g_2 = R_1 R_2 : R_2 \rho_1$

**(D)** $g_1 : g_2 = R_1 \rho_1 : R_2 \rho_2$

**Q.16**

Let $g$ be the acceleration due to gravity at earth's surface and $K$ be the rotational kinetic energy of the earth. Suppose the earth's radius decreases by $2\%$ keeping all other quantities same, then

**(A)** $g$ decreases by $2\%$ and $K$ decreases by $4\%$

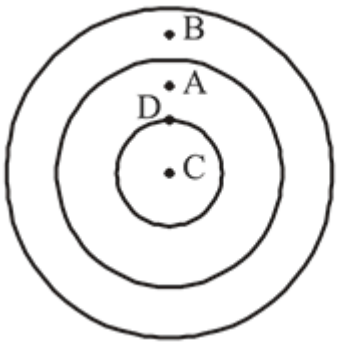
**(B)** $g$ decreases by $4\%$ and $K$ increases by $2\%$

**(C)** $g$ increases by $4\%$ and $K$ increases by $4\%$

**(D)** $g$ decreases by $4\%$ and $K$ increases by $4\%$

**Q.17**

Two concentric spherical shells are as shown in figure. Choose the correct statement given below:




**(A)** Potential at $A$ is greater than $B$

**(B)** Gravitational field at $A$ is less than $B$

**(C)** As one moves from $C$ to $D$ then potential remains constant

**(D)** As one moves from $D$ to $A$ then gravitational field increases

**Q.18**

Intensity of the gravitational field inside the hollow spherical shell is

**(A)** variable

**(B)** minimum

**(C)** maximum

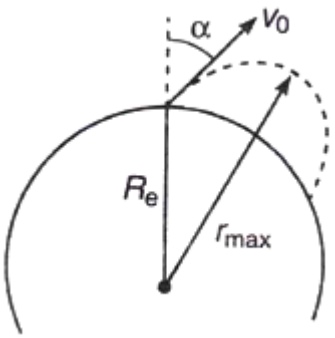
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**(D)** zero

**Q.19**

A projectile of mass $m$ is fired from the surface of the earth at an angle $\alpha = 60^\circ$ from the vertical. The initial speed $v_0$ is equal to $\sqrt{\frac{GM_e}{R_e}}$. The maximum height that projectile can rise from the surface of the earth is $\frac{R_e}{x}$. Find the value of $x$?

$G$ = Universal Gravitational constant
$M_e$ = Mass of earth
$R_e$ = Radius of earth




**(A)** 1

**(B)** 2

**(C)** 3

**(D)** 4

**Q.20**

Two small satellites are moving in circular orbits around the earth at a distance R and R + △ R from the centre of the earth. If their time period of rotation are T and T + △ T respectively, then (ΔR < K)

**(A)** $\Delta T = T\frac{\Delta R}{R}$

**(B)** $\Delta T = 3T\frac{\Delta R}{R}$

**(C)** $\Delta T = \frac{3}{2}T\frac{\Delta R}{R}$

**(D)** $\Delta T = \frac{2}{3}T\frac{\Delta R}{R}$

## Answers & Solutions

**Q.1 Answer:**

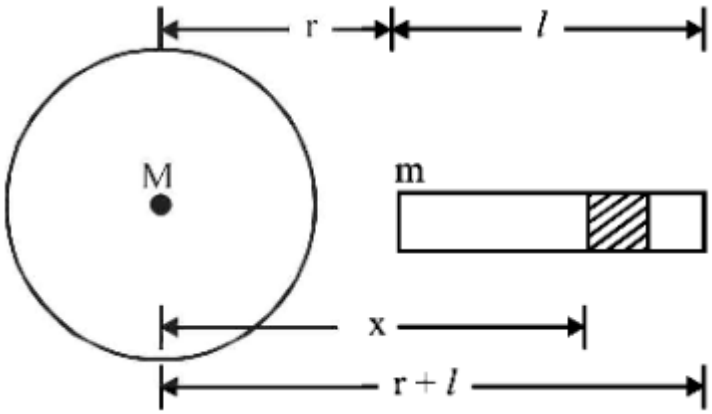
$\frac{GMm}{r(r+l)}$

**Solution:**

**The force of attraction between sphere and shaded position** $dF = GM\frac{\left(\frac{m}{l}dx\right)}{x^2}$

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$F = \int_r^{r+l} \frac{GMm}{lx^2} dx = \frac{GMm}{l} \int_r^{r+l} \frac{1}{x^2} dx$




$= \frac{GMm}{l} \int_r^{r+l} x^{-2} dx = \frac{GMm}{l} \left[\frac{x^{-2+1}}{-2+1}\right]_r^{r+l}$

$= -\frac{GMm}{l} \left[x^{-1}\right]_r^{r+l} = -\frac{GMm}{l} \left[\frac{1}{x}\right]_r^{r+l} = \frac{GMm}{r(r+l)}$

**Q.2 Answer:**

$\frac{7}{8}\frac{GM}{x^2}$

**Solution:**

**Let mass of smaller sphere (which has to be removed) is** $\mathrm{m}$

**Radius** $= \frac{\mathrm{R}}{2}$ **(from figure)**

$\frac{\mathrm{M}}{\frac{4}{3}\pi \mathrm{R}^3} = \frac{\mathrm{m}}{\frac{4}{3}\pi\left(\frac{\mathrm{R}}{2}\right)^3}$

$\Rightarrow \mathrm{m} = \frac{\mathrm{M}}{8}$

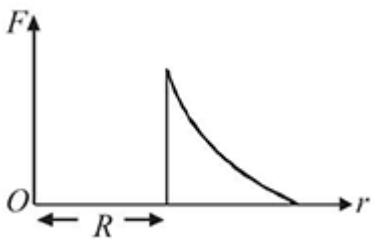
**Mass of the left over part of the sphere**

$\mathrm{M}' = \mathrm{M} - \frac{\mathrm{M}}{8} = \frac{7}{8}\mathrm{M}$

**Therefore gravitational field due to the left over part of the sphere**

$= \frac{\mathrm{GM}'}{\mathrm{x}^2} = \frac{7}{8}\frac{\mathrm{GM}}{\mathrm{x}^2}$

**Q.3 Answer:**




**Solution:**

**The Gravitational field due to a thin spherical shell of radius** $R$ **at distance** $r$**.**

$$F = \frac{GM}{r^2} (\text{If } r > R)$$

**For** $r = R$ **i.e. on the surface of the shell**

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$$F = \frac{GM}{R^2}$$

**Q.4 Answer:**

the gravitational field is necessarily zero

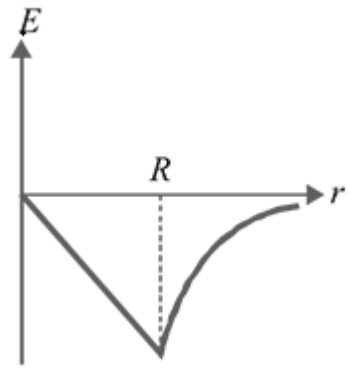
**Solution:**

**As we know that gravitational field and potantial are related as**

$I = -\frac{dv}{dr}$

**if $V = 0$ then gravitational field is necessarily zero, because this would signify to a point at infinity and as we all know gravitational field, which is inversely proportional to $\frac{1}{(\text{distance of the body from the Centre of Mass})^2}$ is also zero at infinity.**

**Q.5 Answer:**




**Solution:**

$\vec{g}' = -\frac{g\vec{r}}{R}$ **for** $r \leq R$ **and** $g' = \frac{g}{(1+r/R)^2}$ **for** $r \geq R$

**so option (a) is correct.**

**Q.6 Answer:**

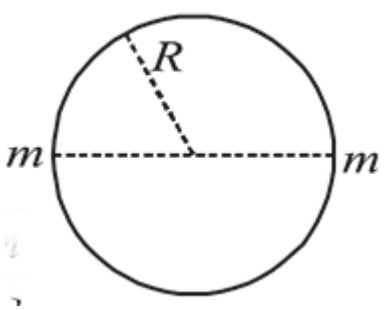
$\sqrt{\frac{Gm}{4R}}$

**Solution:**

**Here, centripetal force will be given by the gravitational force between the two**

**If the velocity of the two particles with respect to the centre of gravity is v then** $v = \omega \mathrm{R}$

$v = \sqrt{\frac{Gm}{4R^3}} \times R = \sqrt{\frac{Gm}{4R}}$

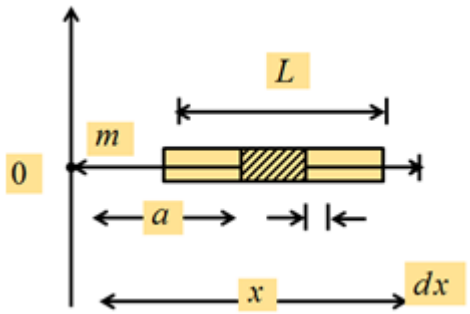



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**Q.7 Answer:**

$Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$

**Solution:**




$\mathrm{dm}=\left(A+Bx^2\right)dx$

$df=\frac{GMdm}{x^2}$

$=F=\int_a^{a+L}\frac{GM}{x^2}\left(A+Bx^2\right)dx$

$=GM\left[-\frac{A}{x}+Bx\right]_a^{a+L}$

$=\mathrm{Gm}\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$

**Q.8 Answer:**

18 m

**Solution:**

**Applying conservation of total mechanical energy principle** $\frac{1}{2}mv^2=mg_Ah_A=mg_Bh_B$

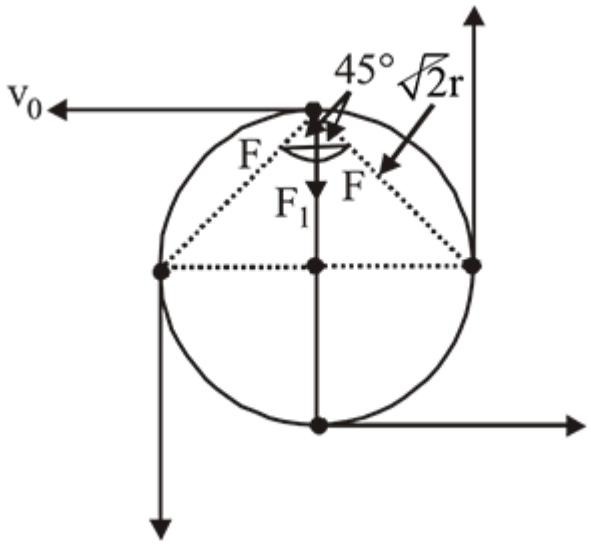
$\Rightarrow g_Ah_A=g_Bh_B$

$\Rightarrow h_B=\left(\frac{g_A}{g_B}\right)h_A=9\times2=18\ \mathrm{m}$

**Q.9 Answer:**

$\left[\frac{\mathrm{Gm}}{\mathrm{r}}\left(\frac{1+2\sqrt{2}}{4}\right)\right]^{\frac{1}{2}}$

**Solution:**




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**Centripetal force = net gravitational force** $\Rightarrow \frac{mv_0^2}{r} = 2F\cos 45^\circ + F_1 = \frac{2Gm^2}{(\sqrt{2})^2}\frac{1}{\sqrt{2}} + \frac{Gm^2}{4r^2}$

$\frac{mv_0^2}{r} = \frac{Gm^2}{4r^2}[2\sqrt{2}+1]$

$\Rightarrow \mathrm{v_0} = \left[\frac{\mathrm{Gm}(2\sqrt{2}+1)}{4\mathrm{r}}\right]^{\frac{1}{2}}$

**Q.10 Answer:**

$\frac{128}{81}\mathrm{G}\pi^2\mathrm{R}^4\rho^2$

**Solution:**

$\mathrm{m_1} = \frac{4}{3}\pi\mathrm{R}^3\rho, \mathrm{m_2} = \frac{4}{3}\pi(2\mathrm{R})^3\rho$, **distance between the centres of the two spherical objects,** $\mathrm{r} = 3\mathrm{R}\ \mathrm{F} = \frac{\mathrm{GM_1\ m_2}}{\mathrm{r}^2}$

$= G\left(\frac{4}{3}\pi R^3\rho\right)\left(8 \times \frac{4}{3}\pi R^3\rho\right) \times \frac{1}{(3R)^2}$

$= \frac{128}{81}\mathrm{G}\pi^2\mathrm{R}^4\rho^2$

**Q.11 Answer:**

$7.5R$

**Solution:**

**Distance between their surfaces** $= 12\mathrm{R} - \mathrm{R} - 2\mathrm{R} = 9\mathrm{R}$

**Since,**

$\mathrm{P} \propto \text{mass}$

$\mathrm{a} \propto \text{mass}$

**Distance** $\propto$ **acceleration**

**So, we can write;**

$\frac{\mathrm{a_1}}{\mathrm{a_2}} = \frac{\mathrm{m}}{\mathrm{jm}} = \frac{\mathrm{s_1}}{\mathrm{s_2}}$

$\frac{s_1}{s_2} = \frac{1}{5}$

$5_{\mathrm{s1}} = \mathrm{s_2}$

**On solving these equations;**

$s_1 = 1.5\mathrm{R}$

$s_2 = 7.5\mathrm{R}$

**Since smaller ball have more acceleration in same time interval, smaller ball will cover more distance.**

**Q.12 Answer:**

$\frac{41\mathrm{GM}^2}{3600\mathrm{R}^2}$

**Solution:**

**Volume of removed sphere**

$V_{\text{remo}} = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{4}{3}\pi R^3\left(\frac{1}{8}\right)$

**Volume of the sphere (remaining)**

$V_{\text{remain}} = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi R^3\left(\frac{1}{8}\right) = \frac{4}{3}\pi R^3\left(\frac{7}{8}\right)$

**Therefore mass of sphere carved and remaining sphere are at respectively** $\frac{1}{8}\mathrm{M}$ **and** $\frac{7}{8}\mathrm{M}$

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**Therefore, gravitational force between these two sphere,** $F = \frac{GMm}{r^2} = \frac{G\frac{7M}{8} \times \frac{1}{8}M}{(3R)^2} = \frac{7}{64 \times 9}\frac{GM^2}{R^2}$ $\simeq \frac{41}{3600}\frac{\mathrm{GM}^2}{\mathrm{R}^2}$

**Q.13 Answer:**

$9.8\ \mathrm{m/sec^2}$

**Solution:**

$g = \frac{GM}{r^2}$. **Since** $\mathrm{M}$ **and** $\mathrm{r}$ **are constant, so** $g = 9.8\ \mathrm{m/s^2}$

**Q.14 Answer:**

$1 : 2$

**Solution:**

$\frac{g_p}{g_e} = \frac{2}{1}; \frac{R_p}{R_e} = \frac{1}{2}$
**So,** $\mathrm{M_p/M_e} = 1 : 2$

**Q.15 Answer:**

$g_1 : g_2 = R_1\rho_1 : R_2\rho_2$

**Solution:**

$g = \frac{4}{3}\pi\rho GR \quad \therefore \frac{g_1}{g_2} = \frac{R_1\rho_1}{R_2\rho_2}$

**Q.16 Answer:**

$g$ increases by $4\%$ and $K$ increases by $4\%$

**Solution:**

$\mathrm{g} = \frac{\mathrm{GM}}{\mathrm{R}^2}$ **or** $\mathrm{g} \propto \mathrm{R}^{-2}$
$\therefore 96$ **change in** $\mathrm{g} = (-2)(\%$ **change in** $\mathrm{R})$
$= (-2)(-2) = +496$
**Rotational kinetic energy,**
$\mathrm{K} = \frac{\mathrm{L}^2}{2\mathrm{I}} \ldots\ldots$ (i)
$\mathrm{L} =$ angular momentum $=$ constant
$\mathrm{I} = \frac{2}{5}\mathrm{mR}^2 \ldots\ldots$ (ii)
**From Eqs. (i) and (ii),**
$\mathrm{K} \propto \mathrm{R}^{-2}$
$\therefore 96$ **change in** $\mathrm{K} = (-2)(96$ **change in** $\mathrm{R}) = (-2)(-2) = +496$

**Q.17 Answer:**

As one moves from $C$ to $D$ then potential remains constant

**Solution:**

**Inside a shell** $V = -\frac{GM}{R}$ **= constant and** $E = 0$
**Outside the shell,** $V = -\frac{GM}{r}$ **and** $E = \frac{GM}{r^2}$ **As** $r$ **increases,** $V$ **increases and** $E$ **decreases.**

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**Q.18 Answer:**

zero

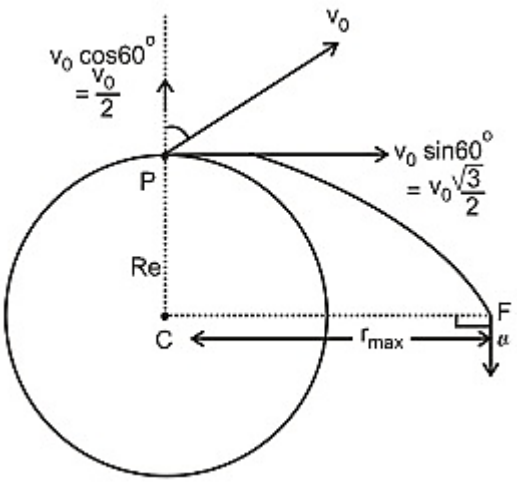
**Solution:**

**At a point inside a spherical shell, the value of gravitational intensity, $I = 0$. If $V = 0$ then gravitational field is necessarily zero.**

**Q.19 Answer:**

2

**Solution:**




**By conservation of angular momentum about C at position P and F**

$$mv_0 \frac{\sqrt{3}}{2} R_e = mv\, r_{max}$$

$$\Rightarrow \quad v = \frac{\sqrt{3} R_e v_0}{2\, r_{max}} \qquad \textbf{...(i)}$$

**By conservation of mechanical energy between P and F**

$$\frac{1}{2} mv_0^2 - \frac{GM_e m}{R_e} = \frac{1}{2} mv^2 - \frac{GM_e m}{r_{max}} \qquad \textbf{...(ii)}$$

**Put (i) in (ii)**

$$\frac{1}{2} mv_0^2 - \frac{GM_e m}{R_e} = \frac{1}{2} m \frac{3R^2 v_0^2}{4\, r_{max}^2} - \frac{GM_e m}{r_{max}}$$

$$v_0 = \sqrt{\frac{GM_e}{R}}$$

$$\Rightarrow \quad \frac{1}{2} m \frac{GM_e}{R_e} - \frac{GM_e m}{R_e} = \frac{3}{8} \frac{mR_e^2}{r_{max}^2} \frac{GM_e}{R} - \frac{GM_e m}{r_{max}}$$

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$$\Rightarrow \quad \frac{1}{2R_e} - \frac{1}{R_e} = \frac{3R}{8\, r_{max}^2} - \frac{1}{r_{max}}$$

$$\Rightarrow \quad \frac{-1}{2R_e} = \frac{3R_e - 8\, r_{max}}{8\, r_{max}^2}$$

$$\Rightarrow \quad 4r_{max}^2 - 8R_e\, r_{max} + 3R_e^2 = 0$$

**Solving the quadratic in $r_{max}$**

**we get**

$$r_{max} = \frac{8R_e \pm \sqrt{16}R_e}{8} = \frac{8R_e \pm 4R_e}{8}$$

$$\Rightarrow \quad r_{max} = \frac{8R_e + 4R_e}{8} = \frac{12R_e}{8} = \frac{3R_e}{2}$$

$$\therefore \quad h_{max} = r_{max} - R_e = \frac{3R_e}{2} - R_e = \frac{R_e}{2}$$

$$\therefore \quad x = 2$$

**Q.20 Answer:**

$\Delta T = \frac{3}{2} T \frac{\Delta R}{R}$

**Solution:**

$$T \propto R^{\frac{3}{2}}$$

$$\frac{(T + \Delta T)}{T} = \frac{R + \Delta R}{R}^{3/2}$$

$$1 + \frac{\Delta T}{T} = 1 + \frac{\Delta R}{R}^{3/2}$$

$$\therefore \Delta R << R$$

$$1 + \frac{\Delta T}{T} = 1 + \frac{3}{2}\frac{\Delta R}{R}$$

$$\frac{\Delta T}{T} = \frac{3}{2}\frac{\Delta R}{R}$$

$$\Delta T = \frac{3}{2} T \cdot \frac{\Delta R}{R}$$

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